Planning Document - Radio Daughterboard

Last Revised: 09/06/2026
Design Lead: Krish Shajpaul, Ethan Huang

Goal

Radio Daughterboard for 2027 Revision of RJ RoboCup Robots. Used in tandem with 2027 Control Board Design Doc to offload networking compute from the main MCU as well provide 2.4/5 Ghz Wifi support through the ESP32-WROOM-1U via esp-hosted. By using a removable module, the current design allows for modular board iteration, rather than requiring a full redesign of the control board itself. Future radio modules simply need to be SPI compatible and they should be able to be controlled with the STM32 main MCU.

Associated Devkit: ESP32-C5-DEVKITC-1-N8R8 Espressif Systems | DigiKey

Requirements

1. Power Input/Output
    a. 3.3 V Rail Input @ 600 mA (maybe 650 mA to be safe)
2. Major Features
    a. ESP32-C5-WROOM-1U-N8R8
    b. Must be able to program using USB-C
    c. Data pins required (as shown on page 5 of ControlBoard_Testing_V0.2.pdf)
        i. Standard Full-Duplex SPI: MOSI/MISO/CLK/CS
        ii. ESP-hosted interrupt/reset pins: Handshake/Data Ready/Reset GPIOs
    d. FXP831.07.0100C Taoglas Limited | RF Antennas | DigiKey
    e. Simple Debug LED to indicate power rail is active
3. Constraints
    a. Connector between control board and radio module must be secure
        i. Do not do simple 2.54mm male/female pin headers; it WILL go loose during comp. Ask me how I know 😐
        ii. As mentioned in the Control Board doc, try to find cheaper Molex alternatives or mezzanine connectors
4. Connectors (pinouts here)
    a. XXX <-> XXX Power
    b. XXX <-> XXX Data1

Critical Parts

Parts that contribute significantly to meeting requirements.

MCU: ESP32-C5-WROOM-1U-N8R8

- Purchase Link: ESP32-C5-WROOM-1U-N8R8 | Digikey

- Datasheet: [Datasheet | Espressif Documentation]
- Design Guide: XXX
- Design Considerations
  - Must XXX